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1. Medical Image Enhancement

- ① Power-law (Gamma) transformation with $\gamma < 1$.
- ② It enhances dark regions without significantly affecting the bright bone regions, making soft tissues more visible.
- ③ The darker tissues become brighter and clearer, while the bones remain visible, improving the overall image quality.

2) Night-Time Surveillance

- ① Log transformation
- ② Log transformation expands low-intensity (dark) pixel values and reveals hidden details, whereas negative transformation only inverts brightness without improving visibility.
- ③ The image will appear brighter, with clearer objects and better visibility of people, vehicles and surroundings.

3) Fingerprint Recognition

① Contrast stretching.

② It increases the contrast between fingerprint ridges and valleys, making the fine details easier to identify.

③ Log transformation would brighten dark regions but may reduce the contrast between ridges and valleys making fingerprint features less distinct than contrast stretching.